3 Introduction to Infinite Series

In the previous chapter, we studied *sequences*, that is, ordered infinite collections of real numbers. A natural question is:

Can we meaningfully add infinitely many real numbers?

For a finite collection of real numbers $a_1, a_2, \ldots, a_n$, the sum

$$a_1 + a_2 + \cdots + a_n$$

is well defined. It is therefore natural to consider the infinite expression

$$\sum_{n=1}^{\infty} a_n.$$

Unlike finite sums, however, infinite sums cannot be manipulated arbitrarily. Without a precise definition, they may lead to contradictions. For example, consider the series

$$1 - 1 + 1 - 1 + \cdots .$$

If we group the terms as

$$(1-1) + (1-1) + (1-1) + \cdots ,$$

we obtain

$$0 + 0 + 0 + \cdots = 0.$$

But if we group them as

$$1 + (-1+1) + (-1+1) + \cdots ,$$

we obtain

$$1 + 0 + 0 + \cdots = 1.$$

To avoid such contradictions, we need a precise definition of the sum of an infinite series. Developing this definition and studying its properties is the main objective of this chapter.

3.1 Convergence and sum of an infinite series

Definition 3.1. *Let (a_n) be a sequence of real numbers. A formal expression of the form*

$$\sum_{n=1}^{\infty} a_n,$$

*or simply $\sum a_n$ or $\sum_n a_n$ is called an **infinite series**.*

Definition 3.2 (Partial Sum)**.** *Let $\sum_{n=1}^{\infty} a_n$ be an infinite series. For each $n \in \mathbb{N}$, the finite sum*

$$s_n := a_1 + a_2 + \cdots + a_n$$

*is called the nth **partial sum** of the series.*

Remark 3.1. *Associated with the series* $\sum_{n=1}^{\infty} a_n$*, we have the sequence of partial sums* (s_n)*, where*

$$s_n = a_1 + a_2 + \cdots + a_n, \qquad n \in \mathbb{N}.$$

Thus, every infinite series naturally determines a sequence of partial sums.

Definition 3.3 (Convergent Series)**.** *An infinite series* $\sum_{n=1}^{\infty} a_n$ *is said to be* ***convergent*** *if the sequence of partial sums* (s_n) *is convergent.*

Remark 3.2. *If* $s_n \longrightarrow s$*, then we call* s *the* ***sum*** *of the series and write* $\sum_{n=1}^{\infty} a_n = s$*.*

Example 3.3. 1. *Consider the series* $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots$ *. Let* $\sum_{n=1}^{\infty} a_n$*, where* $a_n = \frac{1}{n(n+1)}$*. If* $s_n = u_1 + u_2 + \cdots + a_n$*, then*

$$\begin{aligned} s_n &= \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

Hence, $\lim_{n\to\infty} s_n = 1$*. Therefore, the series* $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ *is convergent and its sum is* 1*.*

2. ***(Geometric Series)****Let* $a \in \mathbb{R}$ *satisfy* $|a| < 1$*. Consider the infinite series* $\sum_{n=0}^{\infty} a^n$*. Since the* n*th partial sum is*

$$s_n = \sum_{k=0}^{n} a^k = \frac{1 - a^{n+1}}{1 - a},$$

and $a^{n+1} \longrightarrow 0$*, it follows that* $s_n \longrightarrow \frac{1}{1-a}$*. Thus,* $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \qquad |a| < 1.$

Definition 3.4 (Divergent Series)**.** *An infinite series* $\sum_{n=1}^{\infty} a_n$ *is said to be* ***divergent*** *if the sequence of partial sums* (s_n) *is divergent.*

Example 3.4. 1. *Consider the series* $1 + 2 + 3 + \cdots$ *. Its sequence of partial sums is*

$$s_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Since $\lim_{n\to\infty} s_n = \infty$*, the sequence of partial sums does not converge. Hence, the series* $\sum_{n=1}^{\infty} n$ *is divergent.*

2. ***(Harmonic Series)*** *Consider the series*

$$1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots.$$

Let $\sum\limits_{n=1}^{\infty} a_n$ *be the series, where* $a_n = \frac{1}{n}$. *Let* $s_n = a_1 + a_2 + \cdots + a_n$ *be the sequence of partial sums. Then*

$$\begin{aligned}
s_2 &= 1+\frac{1}{2},\\
s_4 &= 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4} > 1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right) = 1+2\cdot\frac{1}{2},\\
s_8 &= 1+\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right) > 1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right) = 1+3\cdot\frac{1}{2},\\
s_{16} &> 1+4\cdot\frac{1}{2},\\
&\vdots
\end{aligned}$$

and, in general,

$$s_{2^n} > 1+n\cdot\frac{1}{2}.$$

Hence, $\lim\limits_{n\to\infty} s_{2^n} = \infty$. *Since* $s_{n+1} - s_n = a_{n+1} > 0$ *for all* $n \in \mathbb{N}$, *the sequence* $\{s_n\}$ *is monotone increasing. As the subsequence* $\{s_{2^n}\}$ *is unbounded, the sequence* $\{s_n\}$ *is also unbounded. Therefore,* $\sum\limits_{n=1}^{\infty} \frac{1}{n}$ *is divergent.*

3. *(p-**Series**) For* $p > 0$, *consider the series* $\sum\limits_{n=1}^{\infty} \frac{1}{n^p}$. *Then*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if and only if $p > 1$. *Equivalently,*

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \infty, & 0 < p \le 1,\\ \textit{converges}, & p > 1.\end{cases}$$

In particular, the harmonic series $\sum\limits_{n=1}^{\infty} \frac{1}{n}$ *is divergent, whereas the series* $\sum\limits_{n=1}^{\infty} \frac{1}{n^2}$ *is convergent.*

Theorem 3.5. *Let* m *be a natural number. Then the two series* $\sum\limits_{n=1}^{\infty} a_n$ *and* $\sum\limits_{n=m+1}^{\infty} a_n$ *converge or diverge together.*

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$, and $t_n = a_{m+1} + a_{m+2} + \cdots + a_{m+n}$. Then $t_n = s_{m+n} - s_m$, where s_m is a fixed number. If $\{s_n\}$ converges, then $\{t_n\}$ converges, and conversely. If $\{s_n\}$ diverges, then $\{t_n\}$ diverges, and conversely. Hence, both the sequences $\{s_n\}$ and $\{t_n\}$, and consequently the series $\sum\limits_{n=1}^{\infty} a_n$ and $\sum\limits_{n=m+1}^{\infty} a_n$, converge or diverge together. □

Remark 3.6. *The theorem states that removing a finite number of terms from the beginning of a series, or adding a finite number of terms to the beginning of a series, does not affect its convergence or divergence.*

Theorem 3.7 (Algebra of Convergent Series). *Let $\sum\limits_{n=1}^{\infty} a_n$ and $\sum\limits_{n=1}^{\infty} b_n$ be two convergent series with sums A and B, respectively. Then:*

1. *The series $\sum\limits_{n=1}^{\infty}(a_n + b_n)$ is convergent and*

$$\sum_{n=1}^{\infty}(a_n + b_n) = A + B.$$

2. *For every $\lambda \in \mathbb{R}$, the series $\sum\limits_{n=1}^{\infty} \lambda a_n$ is convergent and*

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

Theorem 3.8 (Cauchy Criterion). *The series $\sum\limits_{n=1}^{\infty} a_n$ converges if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that*

$$n, m \geq N \Rightarrow |s_n - s_m| < \epsilon,$$

where (s_n) denotes the sequence of partial sums. Thus, the series $\sum\limits_{n=1}^{\infty} a_n$ converges if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > m \geq N \Rightarrow |a_{m+1} + a_{m+2} + \cdots + a_n| < \epsilon.$$

Proof. The series $\sum\limits_{n=1}^{\infty} a_n$ converges if and only if the sequence (s_n) of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that (s_n) is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately. □

Theorem 3.9. *Suppose that the series $\sum\limits_{n=1}^{\infty} a_n$ converges to s. Then, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \epsilon.$$

Proof. Let $\epsilon > 0$ be given. Since $s_n \to s$, there exists $N \in \mathbb{N}$ such that $|s - s_N| < \epsilon$. Since $\sum\limits_{n=N+1}^{\infty} a_n = s - s_N$, it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \epsilon.$$

This completes the proof. □

Theorem 3.10. *If the series $\sum\limits_{n=1}^{\infty} a_n$ converges, then $\lim\limits_{n\to\infty} a_n = 0$.*

Proof. Let $s_n = \sum\limits_{k=1}^{n} a_k$ be the sequence of partial sums. Since the series converges, (s_n) converges. Hence, by the Cauchy Criterion for sequences, for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \epsilon, \qquad \forall n, m \geq N.$$

Now let $n \geq N + 1$. Then $n - 1 \geq N$, and therefore

$$|a_n| = |s_n - s_{n-1}| < \epsilon.$$

Hence $a_n \to 0$. □

Example 3.11. *Show that the series $\sum\limits_{n=1}^{\infty} a_n$ is divergent, where $a_n = \frac{n}{n+1}$.*

Proof. Since $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$, the necessary condition for the convergence of a series is violated. Hence $\sum\limits_{n=1}^{\infty} a_n$ is divergent. □

Theorem 3.12. *If the series $\sum\limits_{n=1}^{\infty} |a_n|$ converges, then the series $\sum\limits_{n=1}^{\infty} a_n$ also converges.*

Proof. Let $s_n = \sum\limits_{k=1}^{n} a_k$ and $\sigma_n = \sum\limits_{k=1}^{n} |a_k|$ denote the sequences of partial sums of $\sum\limits_{n=1}^{\infty} a_n$ and $\sum\limits_{n=1}^{\infty} |a_n|$, respectively.

Since (σ_n) converges, it is a Cauchy sequence. Thus, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|\sigma_n - \sigma_m| < \epsilon, \qquad \forall n > m \geq N.$$

Then

$$|s_n - s_m| = |a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| = \sigma_n - \sigma_m < \epsilon.$$

Hence, (s_n) is a Cauchy sequence. By the Cauchy Criterion for sequences, (s_n) converges. Therefore, $\sum\limits_{n=1}^{\infty} a_n$ converges. □

Example 3.13. *Show that the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum\limits_{n=1}^{\infty} a_n$, where $a_n = \frac{(-1)^{n+1}}{n}$, is convergent.*

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$ be the sequence of partial sums. Then, for any $p \in \mathbb{N}$,

$$|s_{n+p} - s_n| = \left| \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3} - \cdots + (-1)^{p-1} \frac{1}{n+p} \right| < \frac{1}{n+1}.$$

Let $\varepsilon > 0$. Then $|s_{n+p} - s_n| < \varepsilon$ whenever $n > \frac{1}{\varepsilon} - 1$. Choose $m = \left\lfloor \frac{1}{\varepsilon} - 1 \right\rfloor + 2$. Then m is a natural number and

$$|s_{n+p} - s_n| < \varepsilon$$

for all $n \geq m$ and $p = 1, 2, 3, \ldots$. Hence $\{s_n\}$ is a Cauchy sequence. Since $\mathbb{R}$ is complete, $\{s_n\}$ converges. Therefore, $\sum\limits_{n=1}^{\infty} a_n$ is convergent. □

Exercise 3.1. *1. Consider the constant sequence $a_n = c$, $n \in \mathbb{N}$, where $c \in \mathbb{R}$ is fixed. Determine whether the series $\sum\limits_{n=1}^{\infty} a_n$ is convergent.*

2. Prove that the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is convergent.

3. Find the sum of each of the following infinite series, if it exists.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n + 2^{n+1}}$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{(3n-2)(3n+1)}$.

4. Using the Cauchy criterion, determine whether each of the following series converges.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n}$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$.

3.2 Tests of convergence of a series of positive terms

The convergence or divergence of a series is determined by its sequence of partial sums. Since partial sums are often difficult to compute explicitly, we use *convergence tests*, which determine the nature of a series without evaluating its partial sums directly. We now study some of these tests.

Definition 3.5. *A series $\sum\limits_{n=1}^{\infty} a_n$ is called a* series of positive terms *if $a_n > 0$, for all $n \in \mathbb{N}$.*

Theorem 3.14. *A series of positive real numbers $\sum\limits_{n=1}^{\infty} a_n$ is convergent if and only if the sequence of partial sums $\{s_n\}$, $s_n = \sum\limits_{k=1}^{n} a_k$, is bounded above.*

Remark 3.15. *Since $s_n = a_1 + a_2 + \cdots + a_n$ is a monotone increasing sequence, if it is not bounded above, then $\lim\limits_{n\to\infty} s_n = \infty$. Hence, the series $\sum\limits_{n=1}^{\infty} a_n$ diverges to ∞. Therefore, a series of positive real numbers either converges to a finite real number or diverges to ∞.*

Theorem 3.16 (Comparison Test)**.** *Let $\sum\limits_{n=1}^{\infty} a_n$ and $\sum\limits_{n=1}^{\infty} b_n$ be two series of nonnegative real numbers. Suppose that $a_n \le b_n$ for all $n \in \mathbb{N}$. Then:*

1. *If $\sum\limits_{n=1}^{\infty} b_n$ converges, then $\sum\limits_{n=1}^{\infty} a_n$ also converges.*

2. *If $\sum\limits_{n=1}^{\infty} a_n$ diverges, then $\sum\limits_{n=1}^{\infty} b_n$ also diverges.*